CHANDU PRASAD BHAIRAPU



Applied AI Engineer • GenAI & Agentic AI

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ABOUT ME

AI engineer with an MSc in Artificial Intelligence and 3 years of professional experience. Since 2024 I have built production software that calls LLM APIs: multi-agent pipelines with LangGraph, RAG systems, and the full-stack apps around them (Python, FastAPI, React), shipped with Docker and CI/CD to AWS and Azure.

EXPERIENCE

Associate Generative AI Developer – Agentic Indexing Pipeline Engineer (Energy Sector)

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning an LLM-based knowledge-extraction pipeline end-to-end (data ingestion, chunking, prompt engineering, and indexing with agentic, deterministic, and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Designed and optimized multi-agent extraction with LangGraph (entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) for regulatory and technical documents, driving precision/recall from 33%/45% to 93%/90% against an SAP-system-derived ground truth.
- Engineered a deterministic (code-based) pipeline for a second, high-volume source stream instead of an LLM approach, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Built with Claude Code as standard practice, using my own custom skills and agentic coding workflows, and designed a Model Context Protocol (MCP) query service that gives Claude Code millisecond-latency retrieval from the knowledge graph.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, enabling domain experts to validate and approve knowledge-graph changes.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, LLM APIs, MCP, RAG, GraphRAG, FastAPI, React, Neo4j, MongoDB, Postgres, Azure, Docker, Kubernetes**

GenAI Developer (AI Consulting Intern) – Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed an agentic AI application with LangGraph that orchestrates requests across vector database queries, general inference, and summarization, and deployed it on AWS.
- Built the full stack: React frontend and Python backend with FastAPI, connecting the user interface to the agentic backend; API testing with Postman.
- Containerized the entire stack with Docker and deployed it to AWS using CDK; version-controlled in GitLab with a CI/CD pipeline.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, React, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) – Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated evaluation of RAG generation and retrieval with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Built AWS cloud infrastructure with Lambda, SQS, and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.
- **Tech stack: Python, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, PyTorch, AWS (Lambda, SQS, DynamoDB, S3, CDK), Azure AI Search, Docker, Streamlit**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity) from time-series process data, engineering lag features and rolling statistical windows from sensor data.
- Deployed a Streamlit dashboard letting chemists simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, Streamlit**

Machine Learning Engineer – Electric Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: LLM Gateway | Python | FastAPI | TypeScript | LLM APIs

- Built a provider-agnostic abstraction layer over multiple LLM provider APIs with token budgeting, streaming, retries, and fallback between providers, logging latency, token usage, and cost for every request.
- Built a dashboard tracking KPIs (cost per request, latency percentiles, evaluation pass rate) in React and TypeScript; AI-generated tests (Claude Code), reviewed by hand and run in a CI pipeline.

TECHNICAL SKILLSET

- **AI Engineering:** LLM APIs, prompt engineering, LangGraph, LangChain, multi-agent systems, RAG, GraphRAG, MCP, AI output evaluation (RAGAS, PytrecEval), Hugging Face Transformers
- **AI Coding Assistants:** Claude Code (custom skills, agentic coding workflows)
- **Programming & Full Stack:** Python, SQL, FastAPI, REST APIs, React (JavaScript), Streamlit, Postman
- **Cloud & DevOps:** AWS (CDK, Lambda, EC2, SQS, DynamoDB, S3, VPC), Azure, Docker, Kubernetes, GitLab CI/CD, Git
- **Databases:** SQL (Postgres), NoSQL (MongoDB, DynamoDB), Neo4j, vector databases (Qdrant, Pinecone, Azure AI Search), SPARQL
- **Machine Learning:** Scikit-Learn, PyTorch, TensorFlow, XGBoost, time-series analysis
- **Methodologies:** Agile

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering, Minor in Computer Science & Engineering — Aug 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

CONFERENCE PUBLICATIONS

[1] Navigating the Future: An approach of Autonomous Indoor Vehicles

CERTIFICATIONS

- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

German (B1) | English (Business Fluent) | Hindi (Fluent) | Telugu (Native)